

WEEK 8



Your Final Minigame

Combine everything

지금까지 배운 모든 걸 모아 게임을 완성해요

BIG PICTURE

No new ideas — just build 🎉



Library + chat commands + sensing + events = a playable game.

새 개념 없이, 가진 도구로 완성

HOW TO COUNT

Count the axis — one block, one number



- 1 Stand on the start block — that spot is $(0, 0, 0)$.
- 2 $x \rightarrow$ walk **right**. Add **+1** for every block.
- 3 $y \rightarrow$ jump **up**. Add **+1** for every block.
- 4 $z \rightarrow$ walk **forward**. Add **+1** for every block.

Say them in order every time: $(x, y, z) = \text{across, up, forward}$.

한 칸 움직이면 +1 — 오른쪽 x , 위로 y , 앞으로 z . 순서는 늘 (x, y, z)

TYPE IT IN

Put the numbers into Minecraft



^x
across →

^y
up ↑

^z
forward ↗

```
place STONE at (x, y, z)
```

- ✓ Open **Code Builder** (press **C** in the world)
- ✓ Type the numbers in order — **(x, y, z)**
- ✓ Press **Run ▶** — your block appears at that spot

코드 빌더(C) 열고 → (x, y, z) 순서로 숫자 넣고 → Run ▶

CONCEPT

One command starts the game

```
def on_start():  
    build_arena()  
    player.teleport(room_1)  
    player.say("Reach the gold block!")  
    player.on_chat("start", on_start)
```

Type `start` → build the world, drop the player in, give the goal.

시작 명령 하나로 게임이 열려요

CONCEPT

Build the arena from your library

```
def build_arena():  
    build_room()  
    agent.move(FORWARD, 6)  
    build_room()  
    agent.move(FORWARD, 6)  
    build_room()
```

Three rooms in a row, all from week 5 + 7 functions.

라이브러리 함수로 경기장을 지어요

CONCEPT

Traps and goals — events

```
def on_walk():  
    if agent.inspect(DOWN) == RED_CARPET:  
        agent.place(AIR, DOWN)  
        player.say("Trap!")  
    if agent.inspect(DOWN) == GOLD_BLOCK:  
        player.say("You win!")  
player.on_travelled(on_walk)
```

함정과 목표를 이벤트로 처리해요

WATCH IT!

Walk → check → win or trap

```
below = GOLD_BLOCK
```

```
below = RED_CARPET
```

```
inspect(DOWN) == GOLD_BLOCK
```

```
✓ true → say("You win!")
```

```
✗ false → trap: place(AIR, DOWN)
```

▶ Result

밟은 블록이 결과를 정해요

PREDICT 🤔

What happens at each step?

Predict first — then click reveal

```
def on_walk():  
    if agent.inspect(DOWN) == RED_CARPET:  
        agent.place(AIR, DOWN)  
        player.say("Trap!")  
    if agent.inspect(DOWN) == GOLD_BLOCK:  
        player.say("You win!")  
player.on_travelled(on_walk)  
# player crosses red carpet, then gold
```

Reveal output 🔍 reveal

VOCAB

Words from the whole course

goal what the player must reach to win

trap a hazard that fires on a trigger

reward a prize for reaching the goal

playtest playing your game to find bugs

목표

함정

보상

플레이 테스트



DEBUG

What's wrong with this code?

Goal: gold = win, red carpet = trap. Right now they're swapped.

buggy

```
def on_walk():  
    if agent.inspect(DOWN) == GOLD_BLOCK:  
        player.say("Trap!")  
    if agent.inspect(DOWN) == RED_CARPET:  
        player.say("You win!")
```

The two reactions are on the wrong blocks.



DEBUG

The bug

Bug: the actions are swapped — gold should win, red carpet should trap. Match each to the right block.

fixed

```
def on_walk():  
    if agent.inspect(DOWN) == GOLD_BLOCK:  
        player.say("You win!")  
    if agent.inspect(DOWN) == RED_CARPET:  
        player.say("Trap!")
```

YOUR TURN!

Build your minigame

✓ Plan first: goal, rooms, trap, reward

✓ One `start` command builds it all

✓ Reuse functions from your library

✓ Traps + goal fire from events

✓ Playtest, then send a video



QUIZ

QUESTION 1 OF 3

Gold says "Trap!" and red carpet says "You win!". Fix?

Swap them — gold should win, red carpet should trap.

Add more rooms.

Delete both events.

Rename the function.



QUIZ

QUESTION 2 OF 3

`on_travelled` with no check makes the trap fire at spawn. Why?

Spawn is always a trap.

The world is too small.

The event fires every step — it needs a block check.

Traps can't use events.



QUIZ

QUESTION 3 OF 3

For your **final game**, what counts as 'done'?

A player can start it, every trigger works, and they can finish.

It looks neat in the editor.

The code has no red squiggles.

It builds one room.

QUIZ

Your score

0 / 3



Re-read the slides. You'll get it.

↺ retake



You're a game maker!

Functions + games — course complete.

축하해요! 영어코딩 함수와 게임 코스 완주